

The Practitioner's Chemistry

What is actually inside the bottle, the gel that cures under light, and the one substitution that matters most

Learning Objectives

- Explain the "toxic trio" of traditional polish ingredients and why they are now widely reformulated out
- Explain how gel polish cures under UV/LED light, and what a photoinitiator does
- Explain the critical safety difference between EMA and MMA in acrylic systems
- Evaluate biotin and other marketed nail-strengthening actives against real evidence

The "toxic trio," and why most brands dropped it

For decades, standard nail polish formulas commonly relied on three ingredients now referred to as the toxic trio: formaldehyde (a hardening agent and known irritant/allergen), toluene (a solvent linked to headaches and, with heavy occupational exposure, more serious health effects), and dibutyl phthalate (DBP) (a plasticiser flagged for reproductive health concerns). Consumer pressure and regulatory tightening have pushed the majority of professional brands to reformulate without these three specifically — "3-free," "5-free," and "10-free" labelling refers to progressively longer lists of similarly reformulated-out ingredients. Teach clients this plainly: it is a genuine, verifiable formulation change, not empty marketing language, though it is still worth checking a full ingredient list rather than trusting a number alone.

How gel polish actually cures

Gel polish contains photoinitiators — molecules that remain inert in the bottle but react instantly when exposed to a specific wavelength of UV or LED light, triggering a chemical reaction that hardens the liquid gel into a durable, glossy solid. This is why gel polish does not

air-dry the way traditional polish does, and why an under-cured layer — from a lamp with degraded bulbs or too short a cure time — remains soft, tacky, and prone to lifting.

◆ Figure 4.1 · How Gel Cures



Photoinitiators sit inert in the bottle and react only under the correct light wavelength — full, correct cure time is never optional.

EMA versus MMA: the safety line that matters most

Professional acrylic systems use EMA (ethyl methacrylate) as the core building-block monomer — a substance with a long, well-established professional safety record when used correctly, with adequate ventilation. A small number of unregulated, low-cost products still use MMA (methyl methacrylate) instead — a substance banned or restricted from nail use in numerous jurisdictions because it bonds far too aggressively to the natural nail plate, is a significantly more potent allergen and irritant, and makes safe, damage-free removal essentially impossible without harming the natural nail beneath. If a supplier's acrylic liquid is unusually cheap and has an unusually strong, harsh odour, treat that as a warning sign, not a bargain, and confirm EMA composition before ever using it on a client.

Biotin and marketed "strengtheners," honestly assessed

Biotin supplementation shows measurable nail-strengthening benefit specifically in people with a genuine biotin deficiency — a relatively uncommon condition. In people without a deficiency, the evidence for meaningful improvement is much weaker, despite prominent marketing. Topical "nail strengtheners" containing hydrolysing proteins or calcium can offer modest, temporary surface improvement, but cannot repair genuine structural damage originating at the matrix. Set expectations honestly: the single most effective "strengthening" strategy in nearly every case is reducing mechanical and chemical stress on the nail, not any product added on top of it.

Key Terms

0

[object Object]

1

[object Object]

2

[object Object]

Practical Activity · The Formula Audit

Check the ingredient list of one gel polish and one acrylic liquid in your kit (or a product you use). Confirm whether the polish is free of the toxic trio, and confirm the acrylic liquid is EMA-based. Note anything unclear on the label. **Bring your findings to your next live session.**